BEFORE YOUR SESSION

Aircraft analysis with an AI assistant

You will chat with an AI assistant inside a program called Kiro. Kiro looks like a code editor, with files on the left and a chat panel on the right; you only use the chat panel. You type a question about an example aircraft in plain English. The assistant answers by running five engineering tools on the laptop. Every number comes from one of those runs, and every run is recorded.

You
type a question in the chat panel

→

The assistant
reads it and decides which tools to run, in which order. It can pick the wrong tool; the tools refuse inputs that would give wrong numbers.

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The tool connector
passes each request to one of the five tools below, checks it and records it

TiGL
geometry: export, shape check, spans, areas

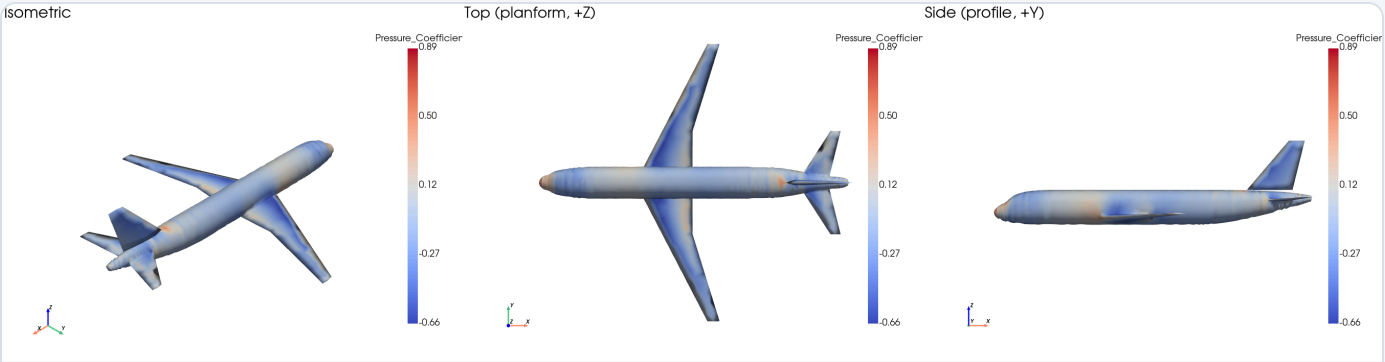
SU2
CFD: lift and drag (Euler, inviscid)

pyCycle
engine: thrust, fuel flow, bypass ratio

NSEG / Aviary
mission: fuel burn, range, weights

OpenAeroStruct
wing-only lattice model, quick trades

All five read and write one shared aircraft file, so each step uses the results of the steps before it.



A real result from this setup: the D150 at Mach 0.78, 2 degrees, 35,000 ft, on the quick laptop mesh. Colours show the pressure coefficient. The assistant produced this in about a minute: geometry about 10 s, CFD about 20 s, picture a few seconds.

The aircraft

The D150, a public 150-seat narrow-body from DLR. Span 33.9 m, reference area 122.4 m². The plain file has the geometry only: no engine and no mass. Some study files also carry stored results from a finer mesh, the D150's published masses and a fixed engine; the tools say when they use them.

The record

Every tool call, with its full input and result, is saved and shown on a report page after the session. The page marks any number in the answer that no tool returned.

HOW TO ASK

Work in steps, one aircraft file per chat

The usual order

- **Geometry first.** Export the geometry; the tool also checks the shape.
- **Then aerodynamics.** CFD gives lift and drag at the Mach number, angle and altitude you state. On a file with stored results, ask for them instead of a new run.
- **Then engine and mission**, if you need fuel or range; they use the results stored in the file.
- **Comparing files** ("which has the larger span?", "what differs between these two designs?") uses a read-only comparison tool and works across files.

Good first questions

Export the geometry of the D150 and tell me its span and reference area.

Run the CFD at the laptop preset at Mach 0.78, 2 degrees and 35,000 ft, and report CL, CD and L/D.

Show the surface-pressure picture of that run.

How long a CFD run takes on a laptop (add about 20 s for the geometry export before the first one)

Request	Mesh	Time	Use it for
"Run the CFD at the laptop preset"	50,000 cells	**about 40 s**	Checking a setup; quick comparisons between cases
"Use the workstation preset"	105,000 cells	**about 2 min**	A finer look at one point
"Use surface density 120"	177,000 cells	**about 2.5 min**	A second or third step of a mesh study
"Use 20 cells across the wing chord" (surface size 0.21 m)	312,000 cells	about 5.5 min	The first step of the mesh study in our paper
The industry preset, or cells smaller than about 0.1 m	1 million cells or more	**over 15 min (estimate)**	Do not start in a session. The assistant is told to ask before it. Meshing alone took 4.5 min at 1 million cells.

Measured on our development laptop (Apple silicon, SU2 on one core), D150, Mach 0.78, 2 degrees, 35,000 ft; your laptop may differ. For scale: in industry, Euler meshes for early design run to a few hundred thousand cells, and production viscous (RANS) meshes run to tens of millions of cells on clusters, hours per case.

Reading the answer

- Each number should name the tool it came from. Ask "which tool gave that?" if it does not.
- The CFD is **inviscid** (Euler): its drag has no skin friction, so lift-to-drag ratios are not cruise values.
- The laptop mesh is a **smoke check**. Lift and drag change as the mesh gets finer; a result needs a mesh study or the stored fine-mesh results.
- A value the tool interpolated between stored runs says so and names the runs.

LIMITS AND RECORDS

What can go wrong, what is out of scope, and where to see what happened

A tool says no

A refusal is the tool doing its job. It says why, and nothing was run.

- **No take-off mass:** the plain file states none; give one, or use a study file that states the D150's 72,500 kg.
- **A second aircraft file:** one file per chat for the analysis tools; start a new chat for another.
- **A faulty shape:** the CFD will not run on a geometry with a shape fault.
- **Mixed flight conditions:** the mission will not combine aero and engine results from different cruise points.
- **Outside the stored range:** the lookup of stored runs will not extrapolate past the angles that were run.

Out of scope in this session

- Changing the geometry (span, sweep, fuselage length) in the CFD model. Variants come as separate files. OpenAeroStruct can vary the wing alone.
- Viscous CFD (RANS) and skin-friction drag in the fuel numbers.
- A real engine: the engine is NASA pyCycle's reference turbofan, sized to a stated design thrust or, failing that, to the drag of the last CFD run; the result says which.
- Wing weight: no tool computes it, so a lighter or heavier wing is not in the fuel numbers.

Errors

- **"The request was throttled by the service":** Kiro's usage limit. Wait a few minutes and ask again.
- **A Docker message during geometry export:** Docker Desktop is not running; start it and ask again.

While it runs

The progress page in the browser shows the current stage, each tool call as it happens and the latest pressure picture. It opens empty for each new chat.

`http://127.0.0.1:8765` (a second chat on the same machine gets the next port)

The one-minute video of what each tool does, from a real run, is shown during the briefing.

Afterwards

"Session reports" on that page lists every session. Each report shows your question, the answer, every tool call with its input and result, and marks any number in the answer that no tool returned.

The facilitator saves the chat and the report after your session; you do not need to save anything.

Rules for the session

- Use only the example aircraft files provided. Do not open other aircraft files.
- What you type, and every tool result the assistant reads, is sent to Kiro's cloud model. Type nothing confidential.
- If the assistant asks to run a command or edit a file, you may say no; the tools do not need it.